



Year 11 Mathematics
Assessment Task 1
Term 1 Week 5, 2013

Name: _____

Teacher: _____

Friday 1st March, 2013

- Answer all questions on the lined paper provided.
- Start each question on a new page.
- Board approved calculators (except graphics calculators) are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- Total possible mark: **40**
- Time allowed: **45 minutes**

Question 1 /10	Question 2 /10	Question 3 /10	Question 4 /10	Total /40
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Question 1 (10 marks)

Marks

(a) Simplify

(i) $\frac{1}{4}mn \times 8n \times (-5p)$ **1**

(ii) $16a^2b \div 28ab^2$ **1**

(iii) $\frac{1}{x+2} + \frac{1}{x-3}$ **2**

(b) Subtract $4x^2 + 7x - 9$ from $5x^2 + 3x - 4$ **2**

(c) Factorise:

(i) $64x^2y^2 - 121$ **1**

(ii) $27x^3 - 125$ **1**

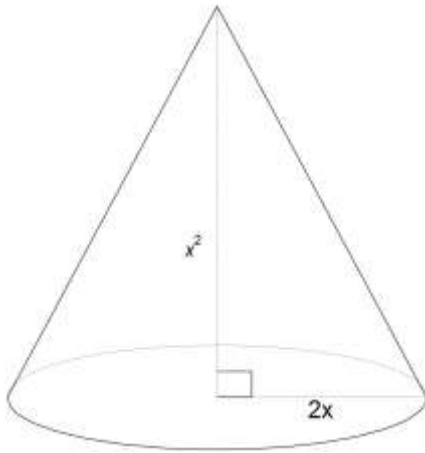
(iii) $3xy + 6x - 5y - 10$ **2**

Question 2 (10 marks)

(a) Make a the subject of the formula $v^2 = u^2 + 2as$. 1

(b) Find the value of b in the formula $a = 6\sqrt{\frac{b}{c}}$ if $a = 42$ and $c = 2$. 2

(c) Find an expression, in terms of x , for the volume of this cone. 2



(d) (i) Factorise $2x^2 + 5x - 52$ 1

(ii) Hence or otherwise, solve $2x^2 + 5x - 52 = 0$ 1

(e) Solve $5a^2 + 4a - 12 = 0$ by using the quadratic formula. 3

Question 3 (10 marks)

(a) Simplify $(3x^{-1})^{-2}$, giving the answer without negative indices. 2

(b) Solve $2 - \frac{x}{8} \leq 6$ 2

(c) Expand and simplify $(\sqrt{11} - 3\sqrt{2})^2$ 2

(d) Rewrite $\frac{2}{3 - \sqrt{7}}$ with a rational denominator in its simplest form. 2

(e) Find the value of x if $\sqrt{54} + \sqrt{24} = \sqrt{x}$ 2

Question 4 (10 marks)

(a) Write 0.000 000 000 192 in scientific notation. 1

(b) Evaluate $\frac{\sqrt{7}}{580(1.02)^{11}}$ correct to three significant figures. 1

(c) Write $x^{\frac{3}{4}}$ in the form $(\sqrt[n]{a})^m$ 1

(d) Solve $16^x = 512$ 2

(e) Write $0.\dot{4}\dot{3}\dot{7}$ as a fraction in simplest form. Show working. 2

(f) Solve simultaneously: 3

$$2x - y + 3z = 3$$

$$3x + 4y - z = -1$$

$$4x - 3y - 2z = 16$$

End of Assessment

(a) (i) $\frac{1}{\cancel{x}} mn \times \cancel{x}^2 \times (-5p) = -10mn^2p$ ①

(ii) $16a^2b \div 28ab^2 = \frac{\cancel{16}^4 \cancel{a}^1 \cancel{b}^1}{\cancel{28}_7 \cancel{a}^1 \cancel{b}^2} = \frac{4a}{7b}$ ①

(iii) $\frac{1}{x+2} + \frac{1}{x-3} = \frac{x-3}{(x+2)(x-3)} + \frac{x+2}{(x-3)(x+2)}$ ①
 $= \frac{\cancel{x}-3 + \cancel{x}+2}{(x+2)(x-3)}$

$\left(\frac{2x-1}{x^2-x-6} \right) = \frac{2x-1}{(x+2)(x-3)}$ ①
acceptable

(b) $5x^2 + 3x - 4 - (4x^2 + 7x - 9)$ ①
 $= \cancel{5x^2} + \cancel{3x} - \cancel{4} - \cancel{4x^2} - \cancel{7x} + \cancel{9}$
 $= x^2 - 4x + 5$ ①

(c) (i) $64x^2y^2 - 121 = (8xy+11)(8xy-11)$ ①

(ii) $27x^3 - 125 = (3x-5)(9x^2+15x+25)$ ①

(iii) $3xy + 6x - 5y - 10 = (3xy+6x) - (5y+10)$
 $= 3x(y+2) - 5(y+2)$ ①
 $= (3x-5)(y+2)$ ①

2.

(a)

$$v^2 = u^2 + 2as$$

$$v^2 - u^2 = 2as$$

$$a = \frac{v^2 - u^2}{2s}$$

①

(b)

$$42 = 6\sqrt{\frac{b}{2}}$$

$$7 = \sqrt{\frac{b}{2}}$$

$$49 = \frac{b}{2}$$

$$b = 98$$

① for substituting the given values into the formula

①

(c)

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi (2x)^2 x^2$$

$$= \frac{1}{3} \pi \times 4x^2 \times x^2$$

$$= \frac{4}{3} \pi x^4$$

① for connect substitution

①

(d) (i)

$$2x^2 + 5x - 52$$

$$= 2x^2 - 8x + 13x - 52$$

$$= (2x^2 - 8x) + (13x - 52)$$

$$= 2x(x - 4) + 13(x - 4)$$

$$= (x - 4)(2x + 13)$$

$$P: -104$$

$$S: 5$$

$$F: 13, -8$$

①

(ii)

$$(x - 4)(2x + 13) = 0$$

$$x = 4, -\frac{13}{2} \text{ or } -6\frac{1}{2}$$

① for both solutions

2.

(e)

$$5x^2 + 4x - 12 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4 \pm \sqrt{4^2 - (4 \times 5 \times -12)}}{2 \times 5}$$

① for correct substitution

$$= \frac{-4 \pm \sqrt{16 - (-240)}}{10}$$

$$= \frac{-4 \pm \sqrt{256}}{10}$$

$$= \frac{-4 \pm 16}{10}$$

$$= \frac{12}{10}, \frac{-20}{10}$$

$$= \frac{6}{5}, -2$$

① for both solutions

3.

$$(a) \quad (3x^{-1})^{-2} = (3)^{-2} (x^{-1})^{-2} \quad (1)$$

$$= \frac{1}{9} \times x^2$$

$$= \frac{x^2}{9} \quad (1)$$

$$\frac{1}{3^2 (x^{-1})^2}$$

or

$$= \frac{1}{9 x^{-2}}$$

$$= \frac{1}{9 \times \frac{1}{x^2}}$$

$$= \frac{1}{\frac{9}{x^2}}$$

$$= \frac{x^2}{9}$$

$$(b) \quad 2 - \frac{x}{8} \leq 6$$

$$- \frac{x}{8} \leq 4 \quad (1)$$

or

$$2 \leq 6 + \frac{x}{8}$$

(1) for
moving
 $\frac{x}{8}$ on RHS

$$-x \leq 32$$

$$-4 \leq \frac{x}{8}$$

$$x \geq -32 \quad (1)$$

$$-32 \leq x$$

$$x \geq -32$$

(1)

$$(c) \quad (\sqrt{11} - 3\sqrt{2})^2 = (\sqrt{11} - 3\sqrt{2})(\sqrt{11} - 3\sqrt{2})$$

$$= 11 - 3\sqrt{22} - 3\sqrt{22} + 18 \quad (1)$$

$$= 29 - 6\sqrt{22} \quad (1)$$

3.

(d)

$$\frac{2}{3-\sqrt{7}} \times \frac{(3+\sqrt{7})}{(3+\sqrt{7})}$$

$$= \frac{6 + 2\sqrt{7}}{9 - 7}$$

$$= \frac{\cancel{6} + 2\sqrt{7}}{\cancel{2}_1}$$

$$= 3 + \sqrt{7}$$

① for multiplying numerator and denominator by the conjugate of $3-\sqrt{7}$

①

(e)

$$\sqrt{54} + \sqrt{24} = \sqrt{x}$$

~~$$\sqrt{144} + \sqrt{144}$$~~

$$\sqrt{9 \times 6} + \sqrt{4 \times 6} = \sqrt{x}$$

$$3\sqrt{6} + 2\sqrt{6} = \sqrt{x}$$

①

$$5\sqrt{6} = \sqrt{x}$$

$$\sqrt{25 \times 6} = \sqrt{x}$$

$$\sqrt{150} = \sqrt{x}$$

$$\therefore x = 150$$

①

4.

$$(a) \quad 1.92 \times 10^{-10} \quad \textcircled{1}$$

$$(b) \quad 0.00367 \quad (\text{or } 3.67 \times 10^{-3}). \quad \textcircled{1}$$

$$(c) \quad x^{\frac{3}{4}} = (\sqrt[4]{x})^3 \quad \textcircled{1}$$

$$(d) \quad 16^x = 512$$

$$2^{4x} = 2^9 \quad \textcircled{1}$$

$$4x = 9$$

$$x = \frac{9}{4} \text{ or } 2\frac{1}{4} \quad \textcircled{1}$$

$$(e) \quad \text{Let } x = 0.4\dot{3}\dot{7}$$

$$10x = 4.\boxed{3\dot{7}}$$

$$100x = 43.\dot{7}\dot{7}$$

$$1000x = 437.\boxed{3\dot{7}}$$

$$1000x - 10x = 437.\dot{3}\dot{7} - 4.\dot{3}\dot{7} \quad \textcircled{1}$$

$$990x = 433$$

$$x = \frac{433}{990} \quad \textcircled{1}$$

$$\therefore 0.4\dot{3}\dot{7} = \frac{433}{990}$$

4.

$$(f) \begin{array}{l} \textcircled{1} \quad 2x - y + 3z = 3 \\ \textcircled{2} \quad 3x + 4y - z = -1 \\ \textcircled{3} \quad 4x - 3y - 2z = 16 \end{array}$$

$$2 \times \textcircled{2} \quad 6x + 8y - 2z = -2$$

$$2 \times \textcircled{2} - \textcircled{3} \quad 2x + 11y = -18$$

$\textcircled{4}$ Elim z

$$3 \times \textcircled{2} \quad 9x + 12y - 3z = -3$$

$\textcircled{1}$ for eliminating the 1st pronumeral

$$\textcircled{1} + 3 \times \textcircled{2} \quad \begin{array}{l} 11x + 11y = 0 \\ x + y = 0 \end{array}$$

$\textcircled{5}$ Elim z

$$2 \times \textcircled{5} \quad 2x + 2y = 0$$

$\textcircled{1}$ for getting a second equation with the same pronumeral eliminated

$$\textcircled{4} - 2 \times \textcircled{5} \quad \begin{array}{l} 9y = -18 \\ y = \frac{-18}{9} \\ y = -2 \end{array}$$

$$\text{Subst into } \textcircled{5} \quad \begin{array}{l} x - 2 = 0 \\ x = 2 \end{array}$$

$$\text{Subst into } \textcircled{2} \quad \begin{array}{l} 3(2) + 4(-2) - z = -1 \\ 6 - 8 - z = -1 \\ -2 - z = -1 \\ -1 = z \end{array}$$

$$\begin{array}{l} x = 2 \\ y = -2 \\ z = -1 \end{array}$$

} $\textcircled{1}$ for getting all 3 pronumerals correct